Medical Image Classification

Training, architecture, dataset composition and validation report

Executive summary

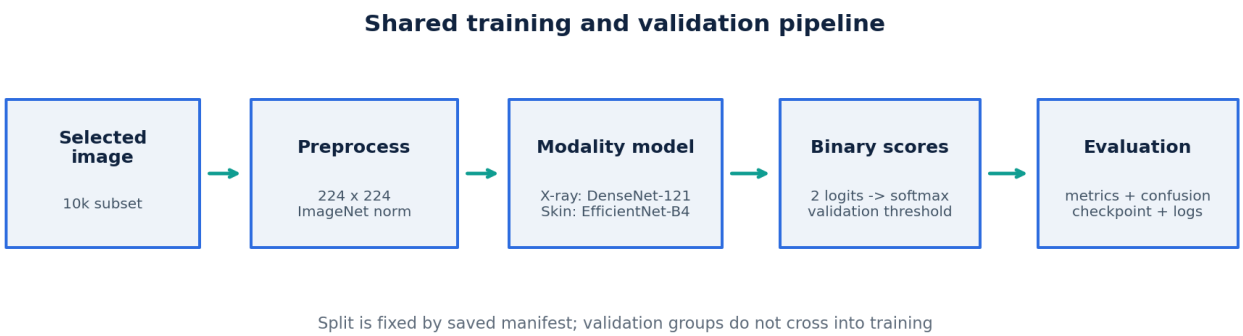
Two pretrained binary image classifiers were trained from the project's NIH ChestX-ray14 and ISIC 2019 data subsets. Each run selected 10,000 unique eligible images, used a fixed seed of 42, and kept its validation split separate from training. The saved checkpoints were reloaded and evaluated on Intel Arc XPU; this report records those measured outputs.

Model	Validation accuracy	ROC-AUC	Positive sensitivity	Positive specificity	Positive F1
DenseNet-121 | Pneumonia	94.8%	0.619	14.7%	97.7%	0.164
EfficientNet-B4 | MEL	70.4%	0.796	75.4%	68.8%	0.562

Interpretation: the skin model separates MEL from NV usefully on this validation split, though its MEL precision is 44.8%. The X-ray model detects only 10 of 68 validation pneumonia cases at its validation-selected threshold (14.7% sensitivity). The prior all-negative failure is improved but remains inadequate for a clinical use claim.

This is a research prototype report. Validation results are not independent test results and do not establish clinical readiness.

1. Model architecture and training



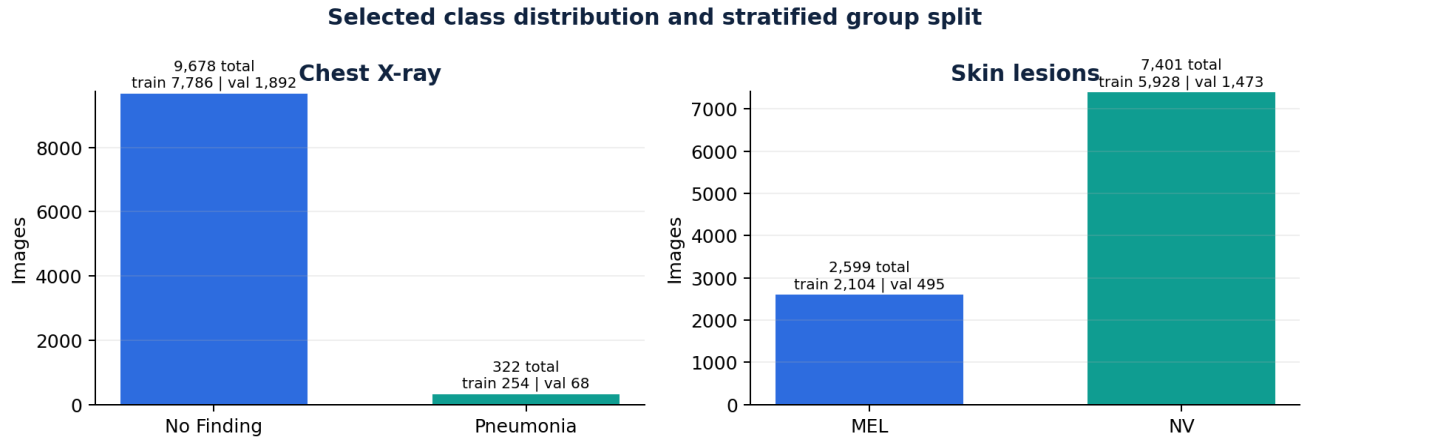
Modality	Backbone and output	Training phases	Decision
Chest X-ray	ImageNet-pretrained DenseNet-121; 2 logits for No Finding and Pneumonia	3 head-only epochs; 1 fine-tuning epoch	Softmax; Pneumonia probability threshold selected on validation (0.2653)
Skin lesion	ImageNet-pretrained EfficientNet-B4; 2 logits for MEL and NV	3 head-only epochs; 1 fine-tuning epoch	Softmax; MEL probability threshold selected on validation (0.4495)

Shared configuration: 224 x 224 input, RGB conversion, ImageNet mean/std normalization, batch size 8, AdamW (learning rate 1e-4, weight decay 1e-4), weighted cross-entropy using inverse-frequency class weights, automatic mixed precision where supported, and 4 DataLoader workers. Pretrained weights were enabled. The current binary experiment uses weighted cross-entropy; the multi-label focal-loss implementation is separate from this binary run.

Runtime selected Intel Arc XPU during the fresh evaluations. The training metadata records 8 logical CPU cores, 4 PyTorch threads and 4 data workers. XPU fallback to CPU remains available in the training pipeline.

2. Dataset selection and split

Selection is capped at 10,000 unique images total per modality, not per class. The NIH subset includes all 322 exact Pneumonia images and samples 9,678 No Finding images without replacement. ISIC sampling preserves the eligible MEL/NV ratio, with no duplication. ISIC grouping uses lesion_id; patient IDs are unavailable in this metadata.

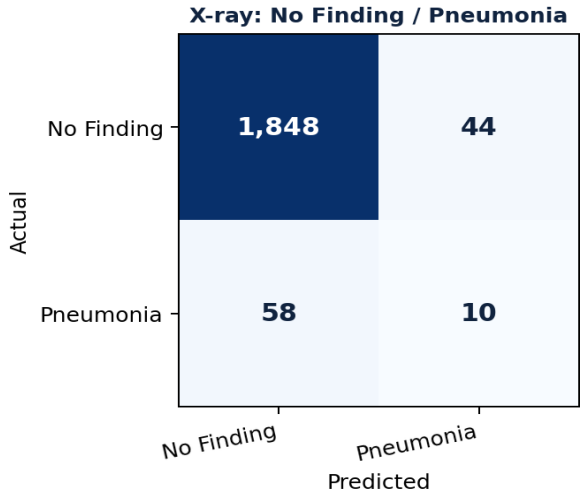


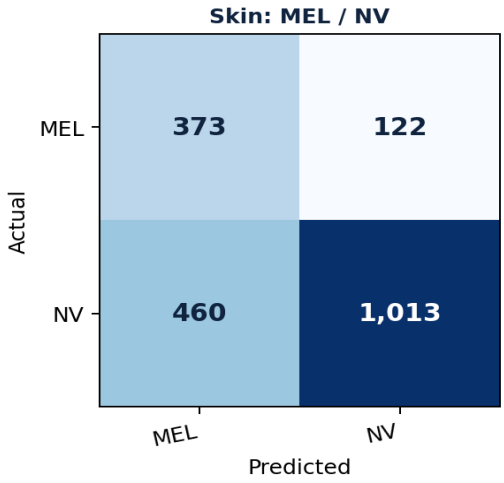
Dataset	Eligible before selection	Selected total	Training split	Validation split	Group handling
NIH ChestX-ray14	60,683 (60,361 No Finding; 322 Pneumonia)	10,000: 9,678 / 322	8,040: 7,786 / 254	1,960: 1,892 / 68	Patient-aware; no group overlap
ISIC 2019	17,397 (12,875 NV; 4,522 MEL)	10,000: 7,401 / 2,599	8,032: 5,928 / 2,104	1,968: 1,473 / 495	Lesion-aware; no group overlap; patient ID absent

X-ray label filter excludes 51,437 other or mixed-label rows; no missing images were recorded in the run metadata. ISIC excludes 7,934 records for other diagnoses; ambiguous and missing-image counts were zero. A stratified group-aware split was saved into each checkpoint manifest and reused for evaluation.

3. Evaluation outputs

The saved best checkpoints were reloaded and evaluated on the exact saved validation manifests. Positive class and threshold are listed per model. Confusion matrices use rows for actual classes and columns for predicted classes.



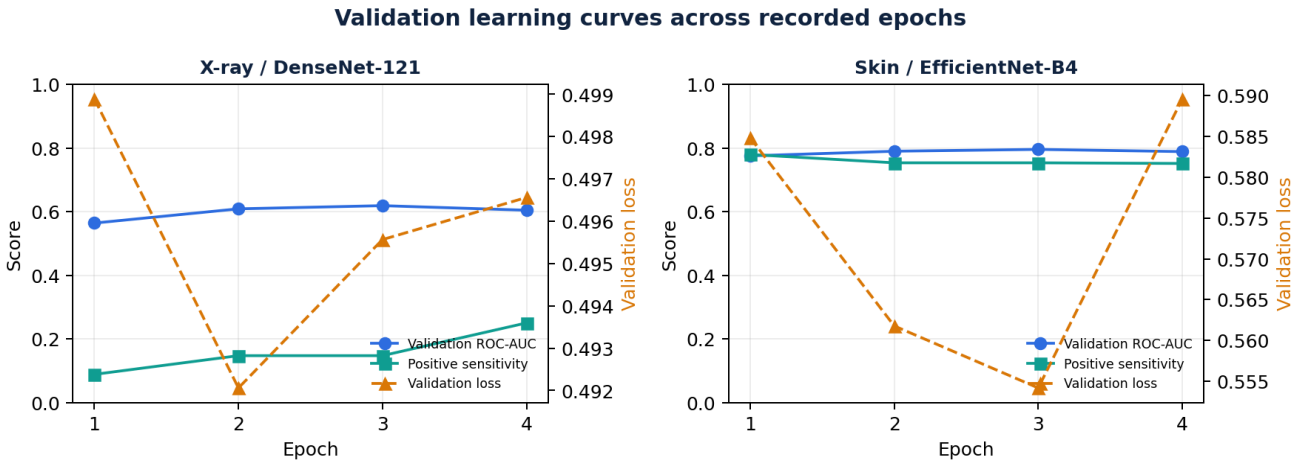


Validation metric	X-ray / Pneumonia	Skin / MEL
Accuracy	94.8%	70.4%
Macro precision	57.7%	67.0%
Macro recall	56.2%	72.1%
Macro F1	56.9%	66.9%
Positive ROC-AUC	61.9%	79.6%
Sensitivity / recall	14.7%	75.4%
Specificity	97.7%	68.8%
Positive precision	18.5%	44.8%
Positive F1	16.4%	56.2%
Threshold	0.2653	0.4495

Confusion matrices

Model (actual rows / predicted columns)	Matrix
X-ray: No Finding, Pneumonia	[[1848, 44], [58, 10]]
Skin: MEL, NV	[[373, 122], [460, 1013]]

4. Learning curves and findings



The plotted curves use the actual epoch records saved during training. The best checkpoints were selected by validation ROC-AUC. Thresholds were selected using validation predictions, so threshold-dependent metrics are descriptive of this same split and are not unbiased test estimates.

What the measurements show

• X-ray: final selected checkpoint validation ROC-AUC is 0.619. Pneumonia sensitivity is 14.7% (10/68); specificity is 97.7%, and positive precision is 18.5%. The model still misses 58/68 pneumonia cases. Overall accuracy is dominated by the much larger No Finding class and should not be used alone.

• Skin: validation ROC-AUC is 0.796. MEL sensitivity is 75.4% (373/495), specificity is 68.8%, and precision is 44.8%. It flags a meaningful number of NV images as MEL (460 false positives), so specialist review remains essential.

• Validation size: 1,960 chest X-rays and 1,968 skin images. No independent test set or external-site validation is reported here. The split results are sensitive to the selected image population and label definitions.

Artifacts and reproducibility

Artifact	Project path
X-ray checkpoint	artifacts/weights/xray_binary_10k_v2.pth
Skin checkpoint	artifacts/weights/skin_binary_10k_v2.pth
Evaluation JSON	artifacts/logs/{xray,skin}_binary_10k_v2_evaluation.json
Training epoch logs	artifacts/logs/{xray,skin}_binary_10k_v2_training.jsonl
Run seed / subset	42 / 10,000 unique selected images per modality

Conclusion

The two-model training workflow, saved checkpoints, and reproducible validation manifests are in place. The skin classifier has moderate validation discrimination. X-ray pneumonia recall remains poor despite improved nonzero sensitivity relative to the earlier all-negative checkpoint. Before any clinical-facing interpretation, the next work should focus on stronger pneumonia recall, patient-level external testing, calibration, and review by qualified clinicians.